

- 4 Dunman High School's PE department intends to develop a Web App for analysing students' NAPFA results. The csv file `napfa_results.csv` contains comma-separated values for each student and their scores across all stations (The sequence of stations is sit ups, standing broad jump, sit and reach, pull-ups, shuttle run and 2.4km run). The sample of the Web App is displayed below:

NAPFA Results

127.0.0.1:5000

Station:

☒ Sit Ups
☐ Standing Broad Jump
☐ Sit And Reach
☐ Pull-ups/Inclined Pull-ups
☐ Shuttle Run
☐ 2.4km Run

Order by:

☒ Ascending
☐ Descending

NAPFA Results:

Student Name	Gender	Sit Ups	Standing Broad Jump	Sit And Reach	Pull-ups/Inclined Pull-ups	Shuttle Run	2.4km Run
Aaron Tan	Female	26	193	50	12	12.33	11.04
Brenda Lim	Male	44	259	20	9	9.1	9.81
Cheryl Ong	Female	43	157	29	19	13.17	14.85
Daniel Lee	Female	36	182	44	13	11.47	12.7
Ethan Ng	Male	50	250	32	6	9.75	12.92
Fiona Chan	Female	42	197	44	7	10.13	12.2
Gabriel Ho	Male	40	241	35	12	11.62	11.82
Hannah Goh	Female	28	188	34	10	12.21	14.15
Isaac Tan	Male	38	230	35	13	9.94	10.29
Jolene Koh	Male	51	209	31	10	9.2	14.91
Kelvin Chua	Female	49	178	55	6	11.85	12.75
Lydia Wong	Male	33	264	29	14	11.92	11.62
Marcus Lim	Female	35	192	34	12	12.9	13.24
Nicole Tan	Female	37	161	43	7	12.08	13.24
Oscar Lee	Male	58	181	48	2	9.6	11.5
Priscilla Ng	Female	25	212	50	10	11.84	13.5
Ryan Ho	Female	42	216	38	17	11.86	15.51
Samantha Ong	Male	42	270	25	4	11.58	12.41
Timothy Chan	Male	45	209	30	5	10.94	12.64
Vanessa Lim	Male	31	185	22	10	10.89	14.48

There are 2 sets of radio buttons available for user selection, specifically Station and Order by. Upon form submission, the results in the html table will be arranged according to the chosen Station in chosen ascending or descending sequence. The sorting is to be done using a **not-in-place sorting algorithm**. Below shows the results when arranged by pull-ups in descending order. [Note: You are **NOT** permitted to use a built-in sorting function]

NAPFA Results:

Student Name	Gender	Sit Ups	Standing Broad Jump	Sit And Reach	Pull-ups/Inclined Pull-ups	Shuttle Run	2.4km Run
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[DHS/2025/Q4 Cont ']

Hint: Sample HTML code for radio button:

```
<input type="radio" name="gender" value="male" checked>Male<br>
<input type="radio" name="gender" value="female">Female<br>
```

<!--checked means it will be auto selected at the start-->

Task 4.1

Write a Python program and the necessary files to create the web application.

[Take note that you are to read the data in the csv file `napfa_results.csv` and store it locally in the flask file.]

Save your Python program as:

`Task_4_<your name>_<centre number>_<index number>.py`

with any additional files/subfolders in a folder named:

`Task_4_web_<your name>_<centre number>_<index number>`

Run the web application.

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Save the webpage output as:

`Task_4_<your name>_<centre number>_<index number>.html`

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Save all your files for Task 4.